Information Security

Muhammad Bilal, Ph.D.

Ph.D. Computer Science and Technology, Zhejiang University, China

MS Computer Science, NTU Faisalabad Pakistan

B.Sc (Hons) Computer Science, UET Lahore Pakistan

GC University Faisalabad

Motivation


فَبِاَيِّ اٰلَآءِ رَبِّكُمَا تُكَذِّبٰنِ
تو (اے جن و انسان!) تم دونوں اپنے رب کی کون کون سی نعمتوں کو جھٹلاؤ گے؟
SURAH REHMAN 55:13

Threats in computer systems

- **Interception:** Gaining unauthorized access to an asset. *Illicit copying, wiretapping,…*
- **Interruption:** Making an asset unavailable/unusable.
  Malicious or accidental destruction of hardware, programs or data, DoS.
- **Modification:** Changing the content or value of an asset. *Alter content of database, modify data in transit,…*
- **Fabrication:** Creation of counterfeit assets.
  Add records to database, insert false messages in network,…

Threats in computer systems

- **Interception:** Interception attacks are any situation where a hacker intercepts and changes communication between two parties without their knowledge.
- **Interruption:** Interruption attacks cause our assets to become unusable or unavailable for our use, on a temporary or permanent basis.
- **Modification:** Changing the content or value of an asset.
- **Fabrication:** Creation of counterfeit assets.

Typical Threats in IT Systems

- **Hardware:**

  Theft, fire, water, dust, smoke, rodents, chemicals,…

  Physical destruction by personnel or outsiders.

  Wiretapping, electromagnetic emanation (Tempest).

- **Software:**

  Unauthorized changes, (un)intentional deletion, wrong version. Logic bombs, hidden side-effects, trapdoors, information leaks. Virus, worms, Trojan horses. Theft, unauthorized copying.

- **Data:**

  Theft, accidental disclosure, inference. Masquerading, unauthorized access.

Jazak Allah Khair